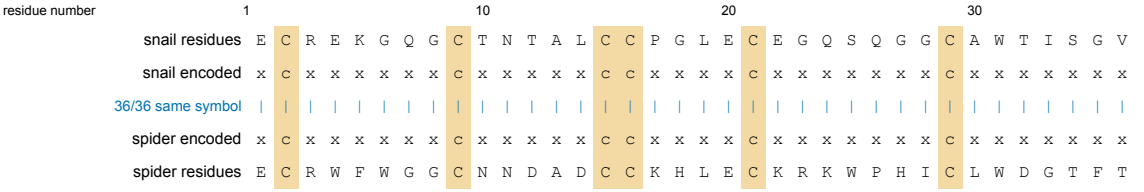
Two toxins sharing the C-C-CC-C-C cysteine framework: cys2 reads them as the same string, cys_hp3 tells them apart and still shows where the cysteines are, and hp_pbotc_1st_ed2 tells them apart but hides the cysteines among the other hydrophobic residues. Conopeptide Mi020 (Conus miles, snail, ConoServer P06751) and Mu/omega-theraphotoxin-Pmu1a (Pterinochilus murinus, spider, UniProt P0DQZ2), 36 residues each, from 5124 mature peptides in ConoServer and Swiss-Prot.

position holding a cysteine in both peptides
the two peptides encode to the same symbol here

a

cys2 c = C, x = the 19 others

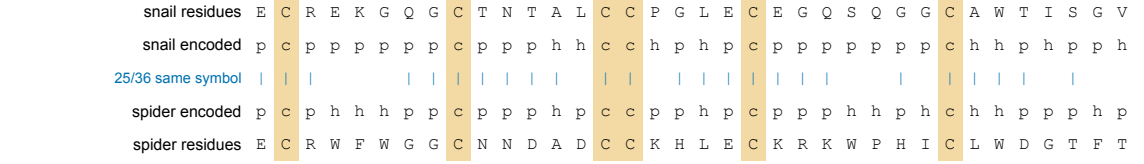


22/22 windows identical at k=15

Every position agrees, so cys2 cannot tell these two peptides apart at all.

b

hp_pbotc_1st_ed_hpc3 (cys_hp3) h = AFILMPVWY, p = DEGHKNQRST, c = C

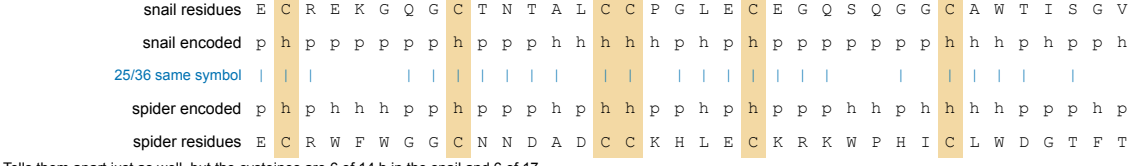


0/22 windows identical at k=15

The six c symbols are the framework; the h and p between them tell the peptides apart.

c

hp_pbotc_1st_ed2 h = ACFILMPVWY (C among them), p = DEGHKNQRST



0/22 windows identical at k=15

Tells them apart just as well, but the cysteines are 6 of 14 h in the snail and 6 of 17 in the spider, so the framework cannot be read off the string.